

Overview

- Understand the structure of vector spaces.
- Identify bases, dimensions and dependencies.
- Use linear combinations as the main tool.

Concept

A vector space allows vectors to be added and multiplied by scalars while respecting mathematical rules.

Attention

A generating set is not always a basis. A basis must also be linearly independent.



This image illustrates the intuition behind the topic. The visual representation helps connect abstract definitions with concrete ideas and provides a quick memory anchor.

Keep this picture as a visual summary of the main concept.

Core concepts

- Linear independence: vectors cannot be expressed as combinations of others.
- Basis: smallest set of vectors generating the whole space.
- Dimension: number of vectors in a basis.
- Span: all possible linear combinations.

Important formula

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

Key condition

A matrix is invertible if and only if $\det(A) \neq 0$.

Quick notes

- Check independence before finding a basis.
- A basis is never unique.
- Dimension stays constant for every basis.
- The determinant gives information about invertibility.